

CHUNGGANG, Korea, Democratic People's Republic ofWMO: **470140**Lat: **41.783N**Lon: **126.883E**Elev: **1086**StdP: **14.13**Time Zone: **9.00 (E09)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.5	-12.9	-20.1	1.9	-14.9	-16.9	2.3	-11.8	12.4	23.4	9.8	18.3	0.0	N/A	0.423

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.0	88.5	75.6	85.8	74.8	83.3	72.9	79.1	86.0	76.8	83.3	74.9	80.9	2.8	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
77.1	147.0	84.1	74.8	135.9	81.6	72.9	127.2	79.4	43.7	86.7	41.3	83.4	39.3	80.9	90.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.2	9.9	7.6	-21.8	93.2	3.2	2.6	-24.1	95.0	-25.9	96.5	-27.7	97.9	-30.0	99.8		
(5)			-21.9	81.9	3.2	3.9	-24.2	84.7	-26.1	86.9	-27.9	89.1	-30.2	91.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	43.0	4.0	12.8	30.1	46.6	59.1	68.2	74.5	72.7	61.8	46.8	28.4	9.5	
	DBStd	25.42	7.98	9.64	9.14	7.66	6.23	4.85	4.10	5.00	6.17	6.94	9.79	8.96	
	HDD50	5297	1426	1041	619	152	8	0	0	0	3	144	649	1255	
	HDD65	8728	1891	1461	1083	554	196	24	1	7	129	564	1098	1720	
	CDD50	2751	0	0	0	49	291	547	759	703	357	45	0	0	
	CDD65	709	0	0	0	0	15	121	295	245	33	0	0	0	
	CDH74	4945	0	0	0	28	331	954	1956	1474	197	5	0	0	
	CDH80	1435	0	0	0	1	72	274	639	431	18	0	0	0	
Wind	WSAvg	1.3	0.6	0.9	1.6	2.4	2.5	1.6	0.9	0.8	1.0	1.4	1.3	0.7	
Precipitation	PrecAvg	33.0	0.4	0.4	0.7	2.0	3.0	4.7	8.0	7.7	3.0	1.4	1.2	0.6	
	PrecMax	48.7	1.1	1.3	1.6	5.2	6.6	8.9	12.5	16.4	6.0	3.2	2.6	1.1	
	PrecMin	25.4	0.1	0.0	0.2	0.3	1.6	1.5	3.0	3.5	0.7	0.2	0.2	0.2	
	PrecStd	5.3	0.3	0.3	0.4	1.1	1.2	1.8	2.8	3.3	1.5	0.8	0.6	0.2	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.0	42.6	60.3	78.5	86.8	90.0	93.3	91.7	82.5	73.7	56.4	35.7	
		MCWB	32.0	36.4	50.2	60.4	67.6	71.6	80.3	80.7	69.6	60.1	47.6	31.9	
	2%	DB	27.1	36.4	54.3	71.7	81.6	86.2	89.0	87.6	79.0	68.1	50.8	31.3	
		MCWB	24.2	32.4	45.1	56.3	64.3	71.1	77.5	77.2	67.7	57.1	44.7	29.1	
	5%	DB	22.4	33.3	48.7	66.5	77.5	83.1	86.2	84.6	76.1	64.0	46.0	27.4	
		MCWB	20.3	29.9	41.0	53.7	62.7	69.5	76.5	75.3	65.8	53.7	41.6	25.2	
	10%	DB	17.8	30.0	43.9	62.3	73.4	80.0	83.5	81.9	72.9	59.7	41.9	24.1	
		MCWB	15.8	27.0	38.0	50.6	60.5	68.7	74.9	73.7	63.5	51.3	38.4	22.1	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	32.4	36.9	51.1	63.3	71.4	76.1	83.0	83.5	72.9	61.5	49.2	32.6
			MCDB	34.5	41.3	59.0	74.1	80.7	83.8	90.1	87.6	78.7	71.0	54.5	34.9
2%		WB	24.3	32.8	45.7	58.3	67.3	73.3	79.9	79.0	69.8	57.8	45.6	29.2	
		MCDB	26.8	35.8	53.3	68.5	77.9	82.5	87.0	85.5	75.9	66.4	50.3	31.2	
5%		WB	20.0	30.1	41.9	54.8	64.5	71.3	77.5	76.8	67.4	54.9	42.1	25.4	
		MCDB	22.3	32.9	47.8	65.0	74.9	80.3	84.5	82.9	73.9	62.6	45.5	27.2	
10%		WB	15.8	26.9	38.4	51.5	61.6	69.3	75.7	74.8	65.2	51.9	38.6	22.2	
		MCDB	17.6	29.9	43.4	60.9	71.3	77.9	81.9	80.4	71.3	58.4	41.6	24.1	
Mean Daily Temperature Range	5% DB	MDBR	23.3	24.8	20.9	23.1	23.6	19.8	16.0	16.6	21.3	22.9	17.1	19.8	
		MCDBR	24.8	24.1	26.2	32.8	32.0	26.4	20.9	20.0	24.3	29.2	21.4	20.6	
		MCWBR	22.4	20.3	18.3	20.0	17.2	13.6	12.1	11.5	14.2	18.5	16.2	18.3	
	5% WB	MCDBR	24.7	23.3	24.9	29.7	28.6	22.7	18.7	17.6	20.6	26.6	20.1	20.1	
		MCWBR	22.5	20.1	18.4	19.9	17.8	14.2	12.3	12.0	13.6	18.3	16.0	18.0	
Clear-Sky Solar Irradiance	taub		0.360	0.425	0.501	0.574	0.616	0.646	0.591	0.523	0.465	0.485	0.427	0.372	
	taud		1.979	1.833	1.749	1.687	1.647	1.658	1.838	2.005	2.074	1.909	1.948	1.981	
	Ebn at Noon		233	234	231	226	220	214	225	237	242	213	206	214	
	Edh at Noon		38	52	64	74	79	78	64	53	46	48	40	36	
All-Sky Solar Radiation	RadAvg		765	1053	1376	1503	1689	1636	1470	1460	1362	1021	671	610	
	RadStd		46	54	71	105	148	168	153	143	128	82	82	44	